Introduction to Responsible Artificial Intelligence (RAI)

Dr. Rabab Ali Abumalloh

Ollscoil Teicneolaíochta an Atlantaigh

Atlantic Technological University

Learning Objectives

Ollscoil Teicneolaíochta an Atlantaigh
Atlantic Technological University

✓ Understand what Responsible AI is and why it matters
✓ Identify where and how AI is used
✓ Explain key ethical principles and risks of AI
✓ Understand AI regulations and governance frameworks
✓ Apply Responsible AI thinking to real-world cases

What is Responsible AI?

Definition
Responsible AI (RAI) refers to:
Designing, developing, and using AI systems in a way that is:

- Fair
- Transparent
- Accountable
- Safe
- Privacy-respecting
- Human-centered

Key Idea:
AI should benefit society and avoid harm.

Responsible AI = More Than Technology

RAI is About Values

Responsible AI is not only about:

- Algorithms
- Data
- Code

It is also about:

- Human values
- Social impact
- Legal responsibility
- Trust

Message:

Technology + Ethics + Society = Responsible AI

AI Application Domains

Where is AI Used?
AI is transforming many sectors:
🏥 Healthcare
💰 Finance
🎓 Education
🚗 Transportation
🛒 Retail
🏭 Industry
📱 Social Media
Example:
AI diagnosing diseases
AI recommending products
AI grading exams

Why is Responsible AI Vital?

Ollscoil Teicneolaíochta an Atlantaigh
Atlantic Technological University

"With Great Power Comes Great Responsibility"
AI can cause harm if misused:

⚠ Biased decisions
⚠ Privacy violations
⚠ Discrimination
⚠ Surveillance misuse
⚠ Dangerous content generation

Example:

- Rejecting loans unfairly
- Predicting crime based on location
- Identifying people from "anonymous" data

Risks of Uncontrolled AI

Ollscoil Teicneolaíochta an Atlantaigh
Atlantic Technological University

Real-World Problems
AI may:
❌ Exclude groups (age, gender, location)
❌ Reinforce stereotypes
❌ Reveal private information
❌ Be used for harmful purposes

Important:
Even “neutral” data can lead to biased results.

Who Sets AI Principles?

Ollscoil Teicneolaíochta an Atlantaigh
Atlantic Technological University

Global Guidelines
Most Responsible AI principles are based on:

👉 Organisation for Economic Co-operation and Development (OECD)

- Founded: 1961
- Members: 38 countries
- Provides global AI policy guidance

Key Principles of Responsible AI (Overview)

According to OECD, Responsible AI must ensure:

- Transparency
- Fairness
- Robustness & Safety
- Privacy
- Accountability
- Inclusiveness & Sustainability

Transparency & Explainability

“Can We Understand AI?”

✓ Users should know:

- How decisions are made
- Why results appear

✗ Not a “black box”

Example:

Loan approval system explains rejection reason.

Fairness & Non-Discrimination

"Equal Treatment for All"

AI must:

✓ Avoid bias

✓ Respect diversity

✓ Treat groups fairly

✗ No discrimination by:

- Gender
- Age
- Race
- Location

Example:

Hiring AI must not prefer one gender.

Robustness & Safety

"Does AI Work Reliably?"

AI systems must:

✓ Be secure

✓ Handle errors

✓ Resist attacks

✓ Work in real conditions

Example:

Self-driving car must react safely to unexpected events.

Privacy & Data Governance

Ollscoil Teicneolaíochta an Atlantaigh
Atlantic Technological University

"Protecting User Data"

AI must:

✓ Respect privacy laws

✓ Protect personal data

✓ Use data responsibly

- No misuse
- No hidden tracking

Example:

Health data must stay confidential.

Accountability

Ollscoil Teicneolaíochta an Atlantaigh
Atlantic Technological University

"Who Is Responsible?"
There must always be:
✓ A human in charge
✓ Legal responsibility
✓ Clear ownership

We cannot say:
"The AI did it, not us."
Developers and organizations are responsible.

Inclusive Growth & Sustainability

Ollscoil Teicneolaíochta an Atlantaigh
Atlantic Technological University

"AI for Everyone"

AI should:

✓ Reduce digital divide

✓ Support social inclusion

✓ Protect environment

✓ Promote sustainable growth

Example:

AI helping disabled users access services.

Who Is Impacted by AI?

Stakeholders
AI affects:

- Individuals
- Communities
- Organizations
- Society
- Governments

Example:
Hiring AI may unfairly reject certain groups.
We must avoid creating new inequalities.

The AI Life Cycle

Three Main Phases
Every AI system goes through:
- Design
- Development
- Deployment

Each phase has unique risks.

Design Phase Risks

Stage 1: Design
In this phase, we define:
✔ Objectives
✔ Business value
✔ Data sources
✔ Budget
✔ Stakeholder support

Possible Risks
⚠ Wrong objectives
⚠ Biased assumptions
⚠ Poor data planning
⚠ Ignoring users
⚠ Ethical blind spots
👉 Bad design = future failure.

Development Phase Risks

Stage 2: Development
In this phase, we:
✓ Collect data
✓ Clean data
✓ Train models
✓ Test performance
✓ Validate results

Possible Risks
⚠ Overfitting
⚠ Underfitting
⚠ Biased training data
⚠ Unrealistic testing
⚠ Data leakage
👉 Models may work in lab but fail in reality.

“Overfitting means the AI memorizes the training data but cannot handle new situations. Underfitting means it never learns properly in the first place.”

Deployment Phase Risks

Stage 3: Deployment
In this phase, AI:
✓ Goes live
✓ Uses real data
✓ Supports decisions
✓ Impacts users

Deployment Phase RisksPossible Risks
⚠ Performance drift
⚠ Unexpected behavior
⚠ Misuse
⚠ Lack of monitoring
⚠ Poor updates
👉 Continuous monitoring is essential.

“Performance drift means the AI slowly becomes worse over time because the real world changes, but the model does not.”

Spectrum of AI Risks (Overview)

Major Risk Categories
AI risks include:

🔐 Security risks
📊 Data risks
🔒 Privacy risks
🏢 Business risks
💰 Financial risks
⚖️ Legal risks
🤝 Trust risks

Security & Operational Risks

Hallucinations
Example:
Avianca Airlines case
Lawyers used AI-generated fake legal cases.
👉 AI invented information.
⚠️ Risk: False confidence.

“AI predicts words based on probability. It does not check facts. So sometimes it produces very confident but completely false information.”

Data Poisoning

Data Poisoning
Example:
Microsoft Tay chatbot
Users trained it to behave offensively.
⚠ Risk: Manipulated models.

“If attackers feed bad or extreme data into the system, the AI learns wrong behavior. So the training data becomes corrupted.”

Data Breaches & Privacy Risks

Data Leakage
Example:
Indian Council of Medical Research breach
815 million records compromised.
⚠ Risk: Legal + reputational damage.

Why Do We Need AI Regulations?

From Risks to Law
We have seen that AI can cause:
⚠ Privacy violations
⚠ Discrimination
⚠ Security failures
⚠ Financial damage
⚠ Loss of trust
These risks affect:
👤 Individuals
🏢 Organizations
🌍 Society

"Different countries have different values, cultures, and priorities.
That's why they regulate AI in different ways."

Global Approaches to AI Regulation

AI regulation varies by region:

- Europe → Rights & privacy focus
- USA → Innovation + safety
- Canada → Ethical AI
- UK → Flexible approach
- China → Strong state control

No single global model yet.

European Union Approach

EU Digital Rights Model
Led by: European Union
Key laws:
✓ GDPR (data protection)
✓ Digital Services Act (DSA)
✓ AI Act (risk-based AI law)
Focus:
Human rights + privacy + accountability

EU AI Act: Introduction

World's First Comprehensive AI Law
Developed by:

- European Commission

The AI Act:

✓ Regulates AI systems
✓ Uses risk classification
✓ Imposes legal obligations
✓ Applies to developers & deployers

Most advanced AI law globally.

EU AI Act: Risk Categories

EU AI Act: Risk Levels
Four Risk Categories
Level
Minimal
- Free use
- Games, spam filters

Limited
- Transparency needed
- Chatbots

High
- Strict controls
- Hiring, exams
- Unacceptable

Banned
- Social scoring

EU AI Act: Examples

EU AI Act: Examples

Risk Levels in Practice

🟢 Minimal Risk

→ AI in video games

🟡 Limited Risk

→ Chatbot must say “I am AI”

🔴 High Risk

→ Student assessment systems

⚫ Unacceptable Risk

→ Mass surveillance systems

Is Regulation Enough?

Ollscoil Teicneolaíochta an Atlantaigh
Atlantic Technological University

From Law to Governance

AI laws:

✓ Set minimum legal requirements

✓ Protect basic rights

✓ Prevent extreme misuse

But laws alone cannot:

✗ Cover every situation

✗ Replace ethical judgment

✗ Manage daily AI operations

Organizations also need internal governance tools.

AI Governance Tools (Overview)

Main Governance Tools
AI governance is built on:

Voluntary Guidelines
Standards
Certifications

Together, they form an AI governance system.

Guidelines tell you what is ethical.
Standards tell you how to build properly.
Certifications prove that you followed the rules.

Voluntary Guidelines (What Are They?)

Ollscoil Teicneolaíochta an Atlantaigh
Atlantic Technological University

Ethical Compasses for AI
Voluntary guidelines:

- ✓ Are not legally binding
- ✓ Provide best practices
- ✓ Support ethical thinking
- ✓ Guide organizations

⚠ Not “one size fits all”
They depend on context and use.

OECD & Global Partnerships

International Guidelines

👉 Organisation for Economic Co-operation and Development (OECD)

Promotes trust, innovation, and human rights.

👉 Global Partnership on Artificial Intelligence (GPAI)

Network of experts using OECD principles.

Focus:

Responsible global AI development.

UN & UNESCO Frameworks

Ollscoil Teicneolaíochta an Atlantaigh
Atlantic Technological University

Human Rights Focus in AI Governance

UNESCO

Provides Ethical AI Recommendations that promote:

- Diversity
- Inclusion
- Human dignity
- Respect for human rights

United Nations (AI Advisory Body)

Develops a global vision for:

- Responsible AI governance
- International cooperation
- Protection of fundamental rights

Risk Management Frameworks

Managing AI Risks: National Institute of Standards and Technology (NIST)
Developed:
AI Risk Management Framework
Helps organizations:
✓ Assign responsibility
✓ Assess risk
✓ Monitor systems

It means identifying problems early, assigning responsibility, and continuously checking whether the system is safe.

AI Standards (What Are They?)

Technical Benchmarks
Standards:
✓ Set quality rules
✓ Ensure consistency
✓ Enable interoperability
✓ Support compliance
Unlike guidelines, they are more technical.

ISO & European Standards

Ollscoil Teicneolaíochta an Atlantaigh
Atlantic Technological University

International Benchmarks
👉 International Organization for Standardization (ISO)

Example:
ISO/IEC 42001 (AI management systems)

ISO/IEC 42001 is the first international standard for managing Artificial Intelligence systems responsibly.
It was developed by the International Organization for Standardization (ISO) and the International Electrotechnical Commission (IEC).

ISO & European Standards

It provides a framework for organizations to:

- ✓ Govern how AI is designed, developed, and used
- ✓ Manage ethical, legal, and technical risks
- ✓ Protect privacy and human rights
- ✓ Ensure transparency and accountability
- ✓ Continuously monitor and improve AI systems

ISO 42001 is like a quality management system for AI.

It helps organizations make sure their AI is safe, ethical, and well controlled.

Reflection: Thinking About Responsible AI

Ollscoil Teicneolaíochta an Atlantaigh
Atlantic Technological University

Take a Moment to Reflect
Think about what you learned today and answer the following:

How does AI affect your daily life or future career?

Which AI risk concerns you the most, and why?

How can you personally contribute to using AI responsibly?

If you were designing an AI system, what ethical principle would you prioritise first?